The following is an important technical result from which several major theorems of calculus easily follow.

Nested Intervals Theorem

Theorem. For each $n \in \mathbb{N}$, assume we are given a closed interval

$$I_n = [a_n, b_n] = \{x \in \mathbb{R} : a_n \leq x \leq b_n\}$$

such that

(i) $a_n \leq a_{n+1}$ and $b_{n+1} \leq b_n$ for all $n \in \mathbb{N}$, so that $I_1 \supseteq I_2 \supseteq I_3 \supseteq I_4 \supseteq \cdots$ (see the picture below).

Then

$$\boxed{\bigcap_{n=1}^{\infty} I_n \neq \emptyset}.$$

If, in addition,

(ii) $\inf\{b_n - a_n\} = 0$,

then

$$\boxed{\bigcap_{n=1}^{\infty} I_n = \{x\}, \text{ where } x = \sup\{a_n\} = \inf\{b_n\}}.$$

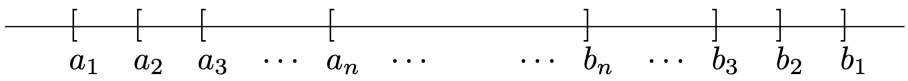


Figure 1: Nested intervals satisfying (i) and (ii).

Proof. Assume (i). Let $A = \{a_n : n \in \mathbb{N}\}$ denote the set of left endpoints and $B = \{b_n : n \in \mathbb{N}\}$ the set of right endpoints.

1. Show that $a_m \leq b_n$ for each $m, n \in \mathbb{N}$. This shows that each b_n is an upper bound for A and each a_n is a lower bound for B.

2. Let $a = \sup A$. Show that $a_n \leq a \leq b_n$ for every $n \in \mathbb{N}$. This shows that $a \in I_n$ for every $n \in \mathbb{N}$, and therefore $a \in \bigcap_{n=1}^{\infty} I_n$. Hence, $\bigcap_{n=1}^{\infty} I_n \neq \emptyset$.

3. Let $b = \inf A$. Show that $a_n \leq b \leq b_n$ for every $n \in \mathbb{N}$. This shows that $b \in I_n$ for every $n \in \mathbb{N}$, hence $b \in \bigcap_{n=1}^{\infty} I_n$.

4. Show that $a \leq b$.

5. Combining inequalities, we have found that $a_n \leq a \leq b \leq b_n$ for all $n \in \mathbb{N}$. Now assume (ii). Show that (ii) implies $a = b$.

6. Show that $\bigcap_{n=1}^{\infty} I_n \subseteq \{a\}$, and therefore that $\bigcap_{n=1}^{\infty} I_n = \{a\}$.

Example: Uncountability of $\mathbb{R}$.

An infinite set S is said to be *countable* if there exists a bijection $f : \mathbb{N} \to S$ (that is, f is a one-to-one function from $\mathbb{N}$ onto S). Otherwise, a S is said to be *uncountable*. This means that for a countable set, its elements can be listed in a sequence, while for an uncountable set there are too many elements to list in a sequence.

We will now apply the Nested Intervals Theorem to prove that $\mathbb{R}$ is uncountable. This has many important and fascinating consequences; for instance, one implication is that there exist real numbers which are *uncomputable* by any computer program.

Let $f : \mathbb{N} \to \mathbb{R}$ be any function. For each n, write $f(n) = x_n$. Construct a sequence of nested closed intervals as follows:

- Let $I_1 = [a_1, b_1]$ be any closed interval not containing x_1.
- Choose I_2 to be a closed subinterval of I_1 which doesn't contain x_2. Explicitly: If $x_2 \notin I_1$, take $I_2 = I_1$. If $x_2 \in I_1 \backslash \{b_1\}$, take $I_2 = [\frac{x_2+b_1}{2}, b_1]$. If $x_2 = b_1$, take $I_1 = [a_1, \frac{a_1+b_1}{2}]$. In any case, we have $I_1 \supseteq I_2$ and $x_2 \notin I_2$. (Drawing a picture of each case might help to convince yourself that we can always find such an interval I_2.)
- Continuing in this way for each x_n for $n \geq 3$, we obtain a nested sequence of intervals

$$I_1 \supseteq I_2 \supseteq I_3 \supseteq I_4 \supseteq I_5 \supseteq \cdots$$

  where $x_n \notin I_n$ for all $n \in \mathbb{N}$. By the Nested Intervals Theorem, $\bigcap_{n=1}^{\infty} I_n \neq \emptyset$.

1. Choose $y \in \bigcap_{n=1}^{\infty} I_n$. Explain why $y \neq x_n$ for all $n \in \mathbb{N}$.

2. Explain why the preceding discussion proves that f is not onto.